

```

1
2 import java.io.FileNotFoundException;
3 import java.io.File;
4 import java.io.*;
5 import java.util.*;
6 import java.text.DecimalFormat;
7 import java.util.List;
8 import java.util.Scanner;
9
10
11 public class AIChatBotGenerator_National
12 {
13     //National Point
14     static Scanner sc = new Scanner(System.in);
15     static ArrayList<Chat_Bots> chat_Bots = new ArrayList<Chat_Bots>(); //SC2
16     static ArrayList<String> languagesComplete = new ArrayList<String>(); //SC1
17
18     public static void main (String args [])
19     {
20         //National Point to use BufferedReader to read this file
21         BufferedReader in = getReader("ExperimentalLanguageBots.txt");//SC3
22
23         Chat_Bots cB = readChatBotPremadeFrameworks(in); //SC3
24
25         chat_Bots.add(cB);
26         while (cB != null)
27         {
28             cB = readChatBotPremadeFrameworks(in); //SC3
29             if(cB != null)
30                 chat_Bots.add(cB);
31         }
32         //National Point to use Scanner to read this file. Use Try/Catch is points
33         try{ //SC10
34             readChatBotLanguages();
35
36         } catch (IOException e){
37             System.out.print("Error reading file Supported_Languages.txt");
38         }
39
40         setChat_Bots(10); //Given
41         ChatBotReporter.chatBotReporter(chat_Bots);
42     }
43
44
45
46
47
48     //Given
49     private static void setChat_Bots(int cB)
50     {
51         Random rand = new Random();
52         Chat_Bots s;
53
54         DecimalFormat df = new DecimalFormat("#.0000");
55         int languageCount = cB;
56         double complexity;
57         int tokenRate;
58         String primaryLanguage; String secondaryLanguage;
59
60         for(int i =0; i<languageCount; i++)
61         {
62
63             do{
64
65                 primaryLanguage = languagesComplete.get(rand.nextInt(languagesComplete.size()));
66                 secondaryLanguage = languagesComplete.get(rand.nextInt(languagesComplete.size()));
67             }while(primaryLanguage.equals(secondaryLanguage));
68             tokenRate = rand.nextInt(10)+11;

```

```

69         double tempComplexity = rand.nextDouble()*10.0 + 1.0;
70         String tempString =df.format(tempComplexity);
71         complexity = Double.parseDouble(tempString);
72         s = new Chat_Bots(primaryLanguage,secondaryLanguage,tokenRate,complexity);
73         chat_Bots.add(s);
74     }
75
76 }
77
78
79
80     //National Points
81     ///////////////////////////////////////////////////NATIONAL METHOD/////////////////////////////////
82     private static Chat_Bots readChatBotPremadeFrameworks(BufferedReader in)
83     {
84         Chat_Bots cB;
85         String primaryLanguage; String secondaryLanguage; int tokenRate; Double complexity;
86         String line ="";
87         String[] data;
88
89         //catches errors in the read but not required since test files will not be corrupted
90         try
91         {
92             line = in.readLine(); //SC4
93
94         }
95         catch (IOException e)
96         {
97             System.out.println("I/O Error");
98             System.exit(0);
99         }
100
101         //ends the read once it gets to the end
102         if (line == null){
103
104             return null;
105         }
106         else
107         {
108             data = line.split(","); //SC5A //" is the delimiter to seperate the fields for the records
109             primaryLanguage = data[0].trim();
110             secondaryLanguage = data[1].trim();
111             tokenRate = Integer.parseInt(data[2].trim()); //SC5B
112             complexity = Double.parseDouble(data[3]); //SC5C
113
114         }
115         cB = new Chat_Bots(primaryLanguage,secondaryLanguage,tokenRate,complexity);
116
117         return cB;
118     }
119
120     //National Points
121     //use this method for Nationals. Must use Scanner and throws IOException (part of the points)
122     public static void readChatBotLanguages()throws IOException
123     {
124         Scanner sc2 = new Scanner(new File("Supported_Languages.txt")); //SC7
125         sc2.useDelimiter(","); //SC8
126
127         while(sc2.hasNext())//SC8
128         {
129             String tempLanguage = sc2.next().trim(); //SC9
130             languagesComplete.add(tempLanguage);
131         }
132     }
133 }
134 //National Points
135 private static BufferedReader getReader(String language)
136 {

```

```

137     BufferedReader in = null;
138     try //SC6
139     {
140         File file = new File(language);
141         in = new BufferedReader(
142             new FileReader(file) );
143     }
144     catch (FileNotFoundException e) //SC6
145     {
146         System.out.println(
147             "The ExperimentallanguageBots.txt file doesn't exist.");
148         System.exit(0);
149     }
150     return in;
151 }
152
153 }
154
155 //////////////////////////////////////
156 //////////////////////////////////////
157 //////////////////////////////////////
158 //National Points. Get Points for Completing all of the required methods.
159 //However, they need to figure out how to make this work with the sort
160 class Chat_Bots
161 {
162     String primaryLanguage;
163     String secondaryLanguage;
164     int tokenRate_Level;
165     double complexity;
166
167     public Chat_Bots()
168     {
169         primaryLanguage = "Esperanto";
170         secondaryLanguage = "Java";
171         tokenRate_Level = 0;
172         complexity = 0.0;
173     }
174
175     public Chat_Bots(String primaryLanguage, String secondaryLanguage, int tR, double complexity)
176     {
177         this.primaryLanguage = primaryLanguage;
178         this.secondaryLanguage = secondaryLanguage;
179         this.tokenRate_Level = tR;
180         this.complexity = complexity;
181     }
182
183
184     public String getDefaultLanguages()
185     {
186         return secondaryLanguage + ", "+primaryLanguage;
187     }
188
189     public String getPrimaryLanguage()
190     {
191         return primaryLanguage;
192     }
193
194     public String getSecondaryLanguage()
195     {
196         return secondaryLanguage;
197     }
198     //Using an ArrayList Sort requires a Double wrapper
199     public Double getComplexity()
200     {
201         return (Double)complexity;
202     }
203
204     public int getTokenRate()

```

```
205 {  
206     return tokenRate_Level;  
207 }  
208  
209  
210 }
```